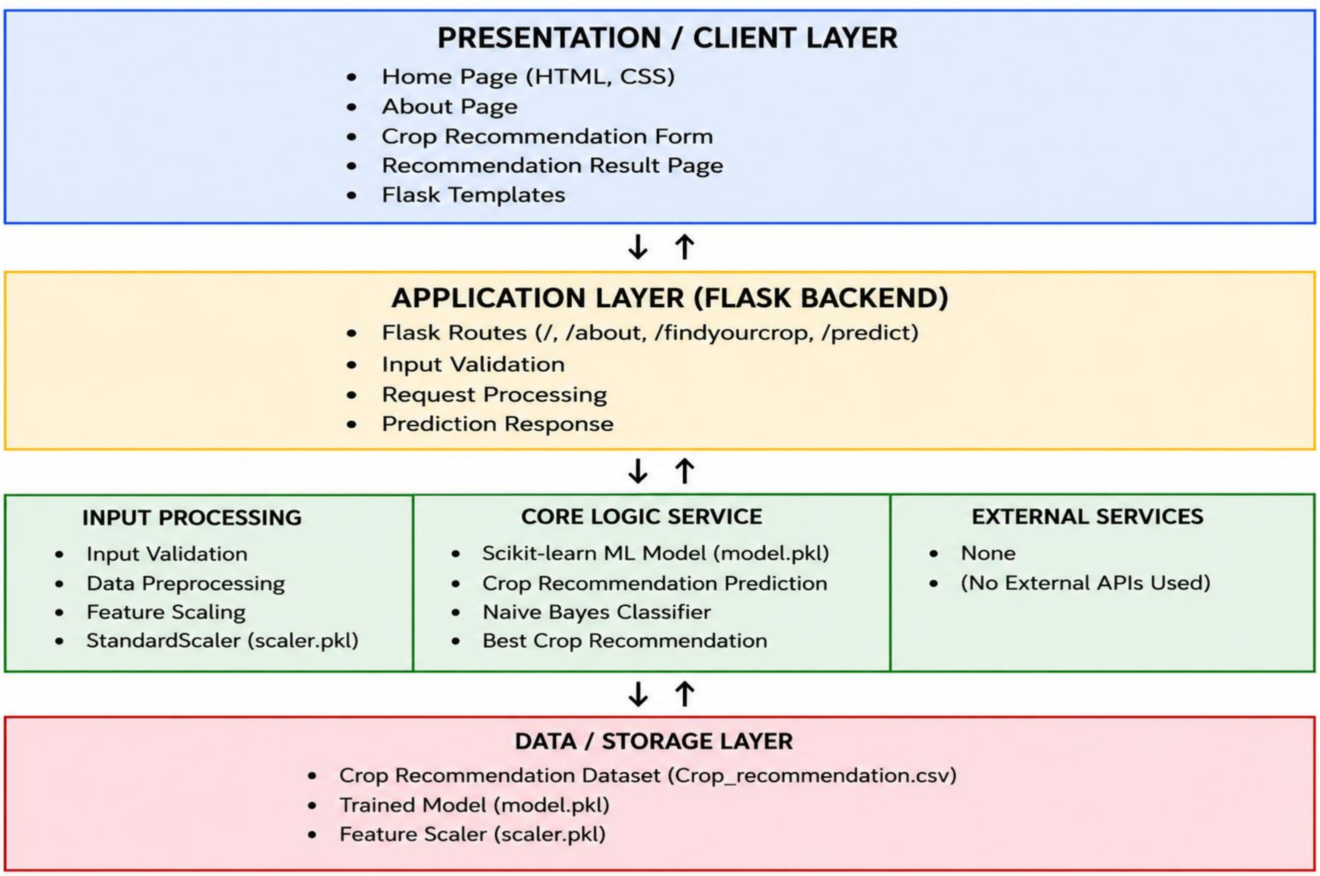
Solution Architecture

Date	27 June 2026
Team ID	xxxxxx
Project Name	OptiCrop – AI-Based Crop Recommendation System
Maximum Marks	5 Marks

Solution Architecture Diagram:

The OptiCrop solution follows a three-tier architecture consisting of the Presentation Layer, Application Layer, and Data Layer. The Presentation Layer provides a user-friendly web interface where farmers enter soil and weather parameters.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Collects soil and weather parameters from the farmer and displays the recommended crop through web pages.	HTML, CSS, Flask Templates
Application Layer (Flask Backend)	Handles routing, input validation, request processing, and communicates with the Machine Learning model.	Python, Flask
Core Logic Service	Preprocesses soil and weather data using the scaler and predicts the most suitable crop using the trained Machine Learning model.	Scikit-learn, Joblib
Data / Storage Layer	Stores the crop recommendation dataset, trained Machine Learning model, and feature scaler required for prediction.	CSV Dataset, model.pkl, scaler.pkl